Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2018
EE 3212 – Broadcast Technologies

Instructions for Candidates:
Answer any **FIVE (05)** questions
All Questions carry equal Marks

No. of questions: **06**
No. of pages : **03**
Time : **(03) Three hours**

01.

i. Explain 'Stereophonic sound' in brief. **(02 Marks)**

ii. What are the advantages and disadvantages of Stereo Recording? **(04 Marks)**

iii. **'Time of arrival and phase differences (Time)'** is one of the techniques used by human ear to detect and locate sounds in the full sphere surrounding. Explain this technique. **(03 Marks)**

iv. Explain 'ZENITH-GE' system for FM Broadcasting and draw the frequency spectrum of the composite FM Stereo signal. **(05 Marks)**

v. Give the following specifications for FM broadcasting in Sri Lanka. **(06 Marks)**

Parameter	Figures and Details
Frequency Band	
Channel Width	
Max Deviation	
Audio Subcarrier Frequency	
Stereo Centre Frequency	
RDS Carrier Frequency	

[Total 20 Marks]

02.

i. Describe 'Pilot Tone'. **(02 Marks)**

ii. Draw the block diagram of VHF FM stereo encoder and explain its operation. **(06 Marks)**

iii. What is the AM modulation technique used in producing the FM Stereo Multiplex (MPX) signal and the reason for using that technique. **(05 Marks)**

iv. Draw the block diagram of a stereo FM receiver indicating three major sections (Mono mode section, Stereophonic mode section and Common section for both Mono and Stereo modes) separately. **(05 Marks)**

v. Briefly explain 'Cross talk between L and R channels'? **(02 Marks)**

[Total 20 Marks]

03.

i. Explain pre-emphasis and de-emphasis. **(03 marks)**

ii. What is the main difference between PAL B and PAL G systems? **(02 Marks)**

iii. State the 'Grassman's law'. **(02 Marks)**

iv. Among colour difference signals (G-Y) is not chosen for transmission. Give reasons for that. **(03 Marks)**

v. Write short notes on following terms. **(2x02 Marks = 04 Marks)**

(a). Additive colour mixing

(b). Quadrature modulation (use to combine colour difference signals into single variable)

vi. Draw a simplified block diagram of monochrome TV camera tube and explain its operation. **(06 Marks)**

[Total 20 Marks]

04.

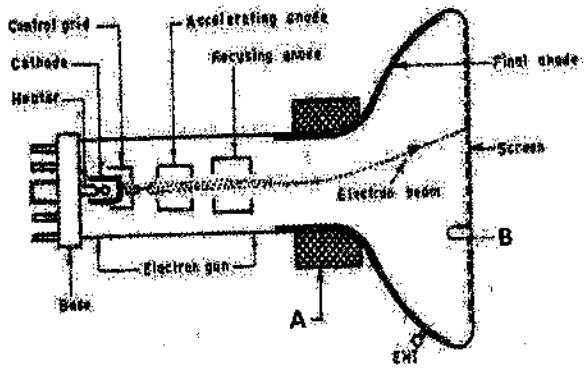
i. Explain how a monochrome TV transmitter works with a suitable diagram. **(05 Marks)**

ii. Draw the simplified block diagram of a Colour Receiver and describe how it works. **(06 Marks)**

iii. What is 'Aspect Ratio' and what are the reasons for selecting it as 4/3? **(04 Marks)**

iv. In monochrome television what are the three main components in Composite Video Signal (CVS) and State the purpose of these three signal components. **(03 Marks)**

v. Name the component parts marked with 'A' and 'B'. **(02 Marks)**



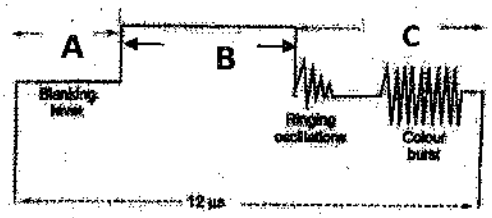
[Total 20 Marks]

05.

i. It is essential that the same co-ordinates be scanned at any instant both at the camera tube target plate and at the raster of picture tube in TV transmission. Explain how this perfect synchronization is ensured in a monochrome television signal transmission. **(04 Marks)**

ii. Explain *'Interlaced scanning'*. **(03 Marks)**

iii. The following figure shows the 'Horizontal Blanking Pulse'. Name A, B and C with their relevant time periods. **(03 Marks)**



iv. Explain the vertical scanning in television system with illustrative diagrams of saw tooth current in the vertical deflection coils mentioning trace & retrace time periods clearly.

(04 Marks)

v. Write short notes on following terms. **(02x3 = 06 Marks)**

(a). Luminance

(b). Shadow mask

[Total 20 Marks]

06.

i. What are the applications of Digital Signal Processing (DSP)? **(03 Marks)**

ii. What are the advantages and disadvantages of DSP systems? **(04 Marks)**

iii. Compare Finite-duration Impulse Response (FIR) filters with Infinite-duration Impulse Response (IIR) filters. **(05 Marks)**

iv. Write short notes on following terms. **(02 x 4 = 08 Marks)**

a) Pulse Code Modulation (PCM)
b) Delta Modulation (DM)
c) Differential Pulse Code Modulation (DPCM)
d) ADPCM: Adaptive Differential Pulse Code Modulation

[Total 20 Marks]